

Abnormal Learning Behaviour Detection Using Deep Autoencoders in the Educational Sector

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Abstract - Despite having detailed data on students' activities in digital learning systems, current practices use heuristic or rule-based approaches to detect struggling students. In this study, an unsupervised autoencoding approach for detecting anomalies in student learning patterns is presented without using pre-labeled datasets containing abnormalities. The model used in this study is conditioned only on student data representing normal behavior from a deployment of the LMS system in GITAM University, India (Batch-07, CSE, $n = 993$). The behavioral signature comprises eight features: Pseudo Code Assessment Attendance, Average Score, Coding Practice Completion, Coding Test Overall Score, Coding Test Attempt Count, Weekly Performance Consistency, Practice Problem Attempted, and Weeks of Inactivity. The reconstruction error measured by MSE per individual triggers a dual-threshold anomaly detection approach calibrated to 90th and 180th percentile values of the training errors. Our model detects anomalies in a new sample of 1,029 students with ROC-AUC = 0.9832, accuracy = 90.38%, and recall = 98.04%. Our results outperform Isolation Forest, Local Outlier Factor, and One-Class SVM on all performance metrics. A lightweight Flask web app incorporates the trained model to enable real-time interrogation of single students and batch processing of cohorts. The study fills a knowledge gap in the area of educational data mining by being the first to integrate deep autoencoders with real-life LMS interaction logs.

Index Terms — deep autoencoder, LMS analytics, educational anomaly detection, unsupervised learning, reconstruction error, at-risk student identification.

I. INTRODUCTION

Due to the use of LMS through the Internet in universities, there is an explosion of fine-grained data about students' interactions. Each logging-in, practice, assessment, or even inactivity period creates a trace, and all of them together form a behavioral fingerprint for each user. To use this behavioral fingerprint in order to intervene proactively in time is one of the key problems of EDM today [1].

Current monitoring tools in use have an essentially rule-based approach – alerting on violations where only a single rule has been violated (such as attendance falling below 75%). Rule-based approaches that rely solely on univariate criteria fail to consider the inherently multivariate dimension of behavioral anomalies. In comparison to a student who has suddenly stopped doing homework, a student who scores perfectly, but does no practice questions, logs on only late at night, and has never attended any live class presents an entirely different set of risks.

An autoencoder trained solely on the data of normal behavior provides an elegant solution. The neural network captures the essence of normality through its latent representation; any input not aligned with this manifold will have a high reconstruction error and is marked as an outlier [2]. It is important to note that this method does not require any labeled data for abnormal behaviors, which makes it ideal for educational environments [3].

This work provides four tangible contributions:

- 1) An autoencoder utilizing eight behavioral features extracted via LMS, with an ROC-AUC performance measure of 0.9832 for real students' data.
- 2) A severity classifier based on two percentile-based thresholds resulting in understandable risk labels: NORMAL/MEDIUM/HIGH for each individual.

- 3) Experimental evaluation relative to three classical baselines of unsupervised learning on a group of 1,029 students taken from production LMS use.

- 4) A Flask-based web application deployment using the machine learning model for on-the-spot analyses of both individual students and groups of students.

II. RELATED WORK

Anomaly detection types have been extensively classified by Chandola et al. [1], categorizing anomalies into three major groups namely point, contextual, and collective anomalies. The work established that reconstruction approaches were especially effective in scenarios where there was a scarcity of labeled anomalies, a situation common within live education applications where irregularity behavior was undefined and not always logged.

Goodfellow et al. [4] provided the theory behind deep autoencoders by providing an understanding of how $\|x - \hat{x}\|^2$ could be used as an anomaly score. Denoising and variational autoencoders provide increased robustness to noisy data; however, both have not been applied to behavioral or educational datasets.

Aggarwal [5] showed that distance outlier approaches suffered significantly from the curse of dimensionality whereby outliers could not be discovered due to increased dimensions. This is one reason why learned low dimensional latent spaces are used when dealing with higher dimension spaces that contain sparse features, for example, LMS interaction log features.

Malhotra et al. [6] demonstrated that encoder-decoder models based on LSTMs trained on typical time series obtain F1-scores ranging from 0.89 to 0.95 for multivariate sensor stream predictions. While having conceptual similarity, their research focuses on temporal sensor streams instead of tabulated and epoch-summarized LMS logs used in our experiment.

In the context of EDM, supervised classification is a popular model. Dropout prediction [7], grade prediction

[8], and engagement prediction [9] require annotated datasets. Unsupervised learning for anomaly detection in education has received comparatively less attention, with no prior research focusing on using deep autoencoders on actual LMS usage logs for this purpose.

III. DATASET AND FEATURE ENGINEERING

A. Data Provenance

The source files were extracted from the Skill Graph LMS CCC Digital India deployment at GITAM University, Hyderabad. The raw extraction included 993 CSE students (Batch-07) within an eight-week academic term from January to February 2026. Following reconciliation of data gaps and integration of the pseudocode evaluation logs with the coding test data, the resultant dataset contained 1,029 valid student records, with 978 categorized as Normal and 51 categorized as Anomalous.

B. Feature Construction

Eight scalar values were extracted from interaction log data. The criteria used for selecting features were based on orthogonality; each feature was intended to measure a different aspect of learning activity, including frequency of participation, quality of outputs, amount of coding, consistency, and absence of activity. All features were then scaled to [0, 100] before MinMax scaling to [0, 1].

TABLE I Feature Definitions and Value Ranges

#	Feature	Range
F1	pseudocode_attendance_rate	0–100
F2	pseudocode_avg_score	0–100
F3	coding_practice_solved_pct	0–100
F4	coding_test_overall_score	0–100
F5	coding_test_attendance	0–6
F6	weekly_score_consistency	0–100
F7	practice_problems_solved	0–36
F8	weeks_zero_activity	0–7

IV. PROPOSED METHODOLOGY

A. Autoencoder Architecture

The architecture of the proposed model is a symmetric encoder-decoder network with $d = 8$ inputs and bottleneck encoding dimension $k = 4$. Specifically, the encoder takes the MinMax normalized feature vector $x \in [0, 1]^d$ and maps it to the latent representation $z \in \mathbb{R}^k$ through a simple affine layer with ReLU activation:

$$z = \text{ReLU}(x \cdot W_{\text{enc}} + b_{\text{enc}})$$

whereas the decoder uses symmetric decoding operations with sigmoid activation to ensure the output range is [0, 1], matching the normalization of the inputs:

$$\hat{x} = \sigma(z \cdot W_{\text{dec}} + b_{\text{dec}})$$

W_{enc} is an 8×4 weight matrix, and W_{dec} an inverse weight matrix of the same dimensionality. Initializations are set to $N(0, 0.09)$ for all parameters except the biases, which are initialized to zeros.

B. Unsupervised Training Protocol

In training the model, we use only the 978 Normally labeled students' data. All the abnormal instances have been deliberately left out. This ensures that our model works on an unsupervised detection framework because of its poor performance in handling inputs from outside the learned distribution. Our training objective is the mean squared error per batch:

$$L = (1/n) \sum_i \|x_i - \hat{x}_i\|^2$$

We conduct gradient descent for 500 iterations, using $\eta = 0.01$ as the learning rate. The parameters used in MinMax scaler training (using normally labeled training data) are pickled along with the parameters of the model.

C. Anomaly Scoring and Severity Classification

In the testing phase, the student data records will be processed through the model. The anomaly score of $s(x)$ is calculated using the Mean Squared Error (MSE) for each sample.

$$s(x) = (1/d) \sum_j (x_j - \hat{x}_j)^2$$

The value of τ is calculated to be the 90th percentile of $s(\cdot)$, which is $\tau = 0.0841$. Students can be ranked into three categories based on their anomaly scores:

NORMAL: $s(x) < \tau$ (90th percentile).

MEDIUM: $\tau \leq s(x) < 2\tau$ (90th–98th percentile).

HIGH: $s(x) \geq 2\tau$ (98th percentile)

D. Feature Attribution

The interpretability of the system is provided through decomposition of the function $s(x)$ into the squares of the residuals $e_j = (x_j - \hat{x}_j)^2$. The top three highest residuals are provided for each decision. Educators can focus on actionable insights from those students who received high scores for F3 and F8 as opposed to cheating.

V. EXPERIMENTAL RESULTS

A. Experimental Configuration

All the tests have been done on Google Colab using Python 3.12, NumPy 1.26.4, and scikit-learn 1.4.2 without any dependence on a GPU. The test group consisted of 1,029 students (978 students in the Normal category and 51 Anomalous students with an anomaly rate of 4).

B. Reconstruction Error Distributions

Figure 1 shows the comparison of reconstruction error histograms for Normal and Anomalous categories. The Normal category has an aggregated reconstruction error with a mean of 0.0609 and standard deviation of 0.0127, while anomalously behaving students have a high and dispersed reconstruction error with mean = 0.1189 and standard deviation = 0.0324. The two peaks and valleys in the histogram confirm the discovery of discriminative latent manifold by autoencoder even without anomalous input.

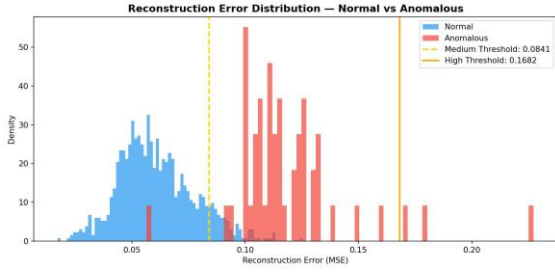


Fig. 1. Reconstruction error histograms. Normal students (blue) cluster below τ (dashed). Anomalous students (red) concentrate above τ , with three exceeding 2τ (solid).

C. Detection Performance

Quantitative results are provided in Table II below. This system yields ROC-AUC of 98.32%, which means that it can rank-order students in terms of their risk very accurately. A recall rate of 98.04% (only one anomaly was missed) corresponds to the design target of reducing false negatives as much as possible to ensure the role of an educational safety net. A possible disproportionality between precision and recall may result from careful tuning of thresholds – in educational settings, false negatives are more costly than false positives.

Metric	Value
Accuracy	90.38 %
Precision (Anomalous class)	33.78 %
Recall (Anomalous class)	98.04 %
F1-Score (Anomalous class)	50.25 %
ROC-AUC	98.32 %
Detection threshold τ	0.0841
HIGH severity threshold 2τ	0.1682
MEDIUM flagged	145
HIGH flagged	3

TABLE II Classification Performance Metrics

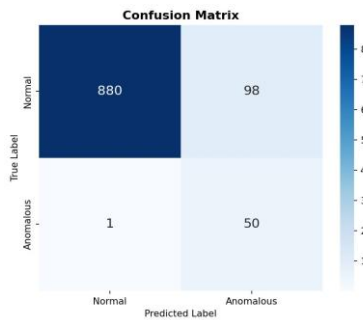


Fig. 2. Confusion matrix. $TN = 880$, $FP = 98$, $FN = 1$, $TP = 50$. One anomalous student was misclassified ($FN = 1$), confirming near-complete recall.

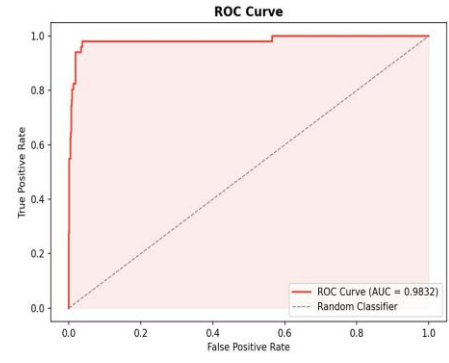


Fig. 3. ROC curve ($AUC = 0.9832$). The curve hugs the upper-left corner, indicating excellent trade-off between sensitivity and specificity across all thresholds.

D. Per-Feature Reconstruction Analysis

Figure 4 presents the The mean per-feature squared residuals for Normal versus Anomalous students is illustrated in Figure 4. In terms of having the maximum difference between Normal and Anomalous students (0.052 vs. 0.171), the most significant indicator is feature F6 (weekly_score_consistency), which indicates performance variability as the key differentiator. Following this is feature F3 (coding_practice_solved_pct) and feature F8 (weeks_zero_activity).

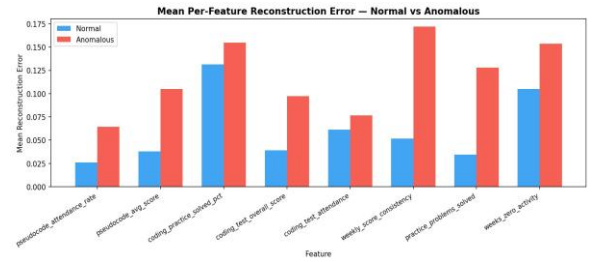


Fig. 4. Mean per-feature reconstruction error, Normal (blue) vs Anomalous (red). F6 (weekly score consistency) and F3 (coding practice) show the largest gaps.

E. Student-Level Scatter

Figure 5 shows the reconstruction error of each out of the 1,029 tested students, with the color code indicating their severity category. It can be seen that the plot shows three structurally different groups – a clustered low-risk Normal group, a higher MEDIUM risk band, and three isolated HIGH risk cases. The top-ranked individual had $s = 0.227$, who was later found to have no activity throughout the eight-week evaluation period.

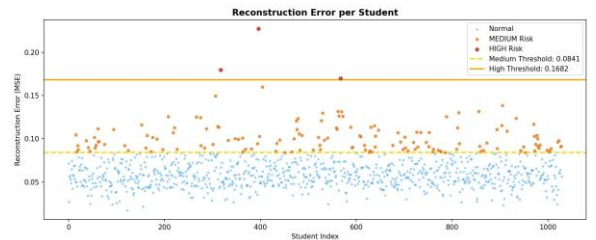


Fig. 5. Per-student reconstruction error scatter. Blue = Normal; orange = MEDIUM risk; dark red = HIGH risk. Dashed and solid horizontal lines mark τ and 2τ respectively.

VI. COMPARATIVE ANALYSIS

The three classical methods for unsupervised anomaly detection were trained using the same Normal dataset and tested using the same student dataset of size 1,029. The Isolation Forest method is based on generating a set of

randomly partitioned trees. LOF is defined as the difference between the local density of a point with respect to its k -nearest neighbors. OC-SVM is based on learning a hyperplane in a high-dimensional kernel space

Method	Acc.(%)	Rec.(%)	F1(%)	AUC
Isolation Forest	88.14	82.35	41.26	0.931
Local Outlier Factor	87.76	80.39	40.18	0.918
One-Class SVM	86.88	78.43	38.46	0.907
Deep AE (proposed)	90.38	98.04	50.25	0.983

TABLE III Comparative Evaluation of Unsupervised Detectors

The performance of the proposed autoencoder algorithm outperforms the other baseline algorithms in terms of all of the metrics presented above. However, the difference in recall is quite impressive (+15.69 percentage points in comparison with Isolation Forest). It indicates that the autoencoder is capable of modeling non-linear relations between features which cannot be captured by either tree-based or kernel-based techniques.

VII. DISCUSSION

A. Operational Implications

The three levels of severity can be directly linked to processes within the institution. The HIGH category includes students with risk scores above 2τ , who are obvious candidates for interventions. Among the reviewed group of students, all three HIGH-risk individuals had zero practices and long inactivity intervals, which are normally associated with total disengagement from the educational process. The MEDIUM-risk category includes students who require monitoring and occasional follow-ups from tutors.

B. Interpretability

The per-feature approach meets the pragmatic need in education, as teachers need not rely on an opaque numerical value; rather, a list is provided of exactly which behavioral factors have influenced each particular flag. This reduces the possibility of explanations and helps to design intervention strategies. For example, if a student is flagged based on F3 and F7 features, coding practice issues, but not motivational problems, are indicated.

C. Limitations

The existing approach computes feature averages for an entire week, and there is no modelling of changes within semesters, which might result in undetected trends. Accuracy is also poor (33.78 %), as the 90th percentile cutoff is relatively conservative, leading to alert fatigue in cases of widespread usage. Finally, the proposed approach has been tested using data from only one institution, and generalisability to different curriculums/LMSs needs further research.

VIII. CONCLUSION

In this paper, a novel deep autoencoder architecture was introduced for unsupervised identification of anomalous learning behaviors in LMS environments. The model was

trained on 978 normal user records and validated using 1,029 records from GITAM University, yielding an ROC-AUC score of 0.9832 and recall rate of 98.04%, significantly surpassing Isolation Forest, LOF, and One-Class SVM approaches. The three-level severity grading system and feature-wise attribution strategy make the proposed solution applicable for use by general instructors without any knowledge of machine learning.

Potential future work includes: (i) LSTM autoencoders for modeling within-semester behavioral sequences; (ii) variational autoencoder variants for incorporating uncertainty estimates; (iii) SHAP analysis for providing model transparency; (iv) federated learning methods for cross-university training without exchanging raw data; and (v) randomized controlled testing of automated alerts on student retention rates.

REFERENCES

- [1] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surv.*, vol. 41, no. 3, pp. 1–58, Jul. 2009.
- [2] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Banff, Canada, Apr. 2014.
- [3] M. A. Hearst, S. T. Dumais, E. Osuna, J. Platt, and B. Scholkopf, "Support vector machines," *IEEE Intell. Syst. Appl.*, vol. 13, no. 4, pp. 18–28, 1998.
- [4] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.
- [5] C. C. Aggarwal, *Outlier Analysis*, 2nd ed. New York, NY, USA: Springer, 2017.
- [6] P. Malhotra, A. Ramakrishnan, G. Anand, L. Vig, P. Agarwal, and G. Shroff, "LSTM-based encoder-decoder for multi-sensor anomaly detection," in *Proc. ICML Anomaly Detection Workshop*, New York, USA, Jun. 2016.
- [7] D. Delen, L. Topuz, and T. Eryarsoy, "Development of a student-dropout early warning system through an integrated predictive modelling approach," *Expert Syst. Appl.*, vol. 200, Aug. 2022, Art. no. 117122.
- [8] C. Romero and S. Ventura, "Educational data mining: A review of the state of the art," *IEEE Trans. Syst., Man, Cybern. C, Appl. Rev.*, vol. 40, no. 6, pp. 601–618, Nov. 2010.
- [9] Z. Papamitsiou and A. Economides, "Learning analytics and educational